RESEARCH OVERVIEW · NOVEMBER 2026

Anatomy as a bridge

How much anatomical information is available, and how should a model use it?

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THREE DIRECTIONS

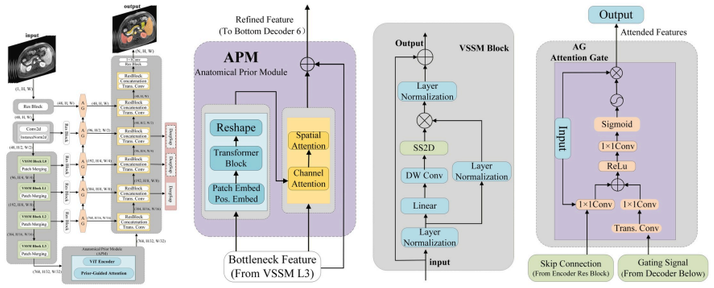
01	UNDERSTAND	Segment anatomy across organs and whichever MRI sequences exist
02	RECOVER	Synthesize missing MRI, diffusion maps or 3D CT from limited inputs
03	REVIEW	Check whether an image retains the anatomy a report refers to

6 studies | **3** accepted, all oral | **3** under review | **5** as first author

01 UNDERSTAND · First author · **EMBC 2026** ORAL

AP-Mamba

Anatomy-Aware State Space Models for Multi-organ Abdominal MRI Segmentation



A visual state-space network with an Anatomical Prior Module for global organ layout and co-occurrence; attention gates refine boundaries.

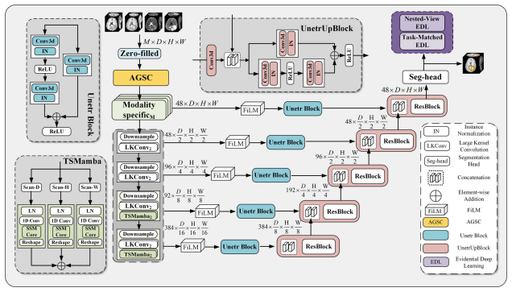
Consistent Dice and NSD gains over CNN, Transformer and recent Mamba baselines.

CHAOS-MRI (5-fold CV) · AMOS-MRI — **Paper** **Code**

01 UNDERSTAND · First author · **IEEE TMI** Under review

ReLiSS

Modality-Adaptive Representation Conditioning for Multimodal MRI Segmentation under Arbitrary Modality Availability



Availability-gated spatial–spectral conditioning adapts one CNN–Mamba segmenter to any subset of MRI sequences, with evidential uncertainty.

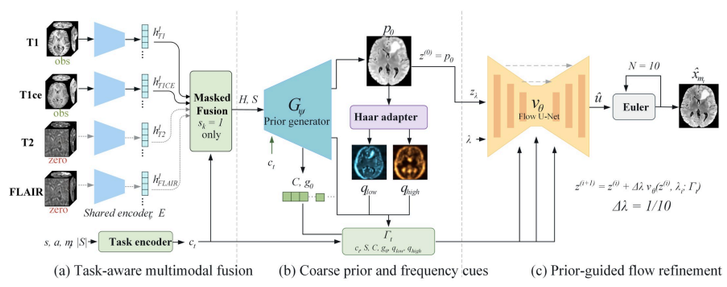
Highest mean Dice under incomplete inputs: **80.82%** BraTS2021, **65.08%** ISLES2022, **70.68%** WMH2017, **64.20%** TSC.

28 incomplete-modality settings — **Paper** **Code**

02 RECOVER · First author · **BIBM 2026** ORAL

PriorFM

Prior-Guided Flow Matching for Missing Modality MRI Synthesis



Fuses any subset of observed sequences into a coarse anatomical prior, then refines it by conditional flow matching in a few Euler steps.

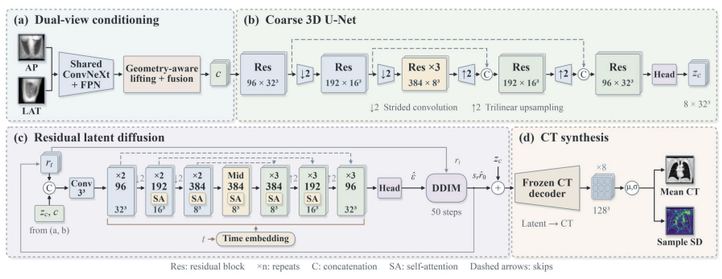
Best average synthesis over **32** source-to-target tasks; synthesized scans give segmentation Dice close to real scans.

BraTS2021 · CHAOS · Diff5T — **Paper** **Code**

02 RECOVER · First author · **ISBI 2027** Under review

UPath-X2CT

Geometry-Aware Biplanar Chest CT Reconstruction with Residual Latent Diffusion



Calibrated projections lift biplanar X-ray features onto a shared 3D grid; conditional diffusion refines the residual of a coarse latent CT.

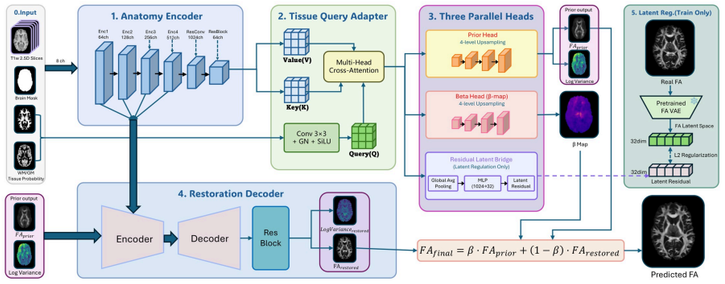
Outperforms five baselines in PSNR, SSIM and MAE, with **+1.23 to 1.80 dB** PSNR over DVG-Diffusion.

LIDC-IDRI · MSD Lung · CT-RATE — **Paper** **Code**

02 RECOVER · Third author · **BHI 2026** ORAL

ATLANTIS

An Anatomy-guided Tissue-Latent Atlas Network for DWI-derived White-Matter Anisotropy Synthesis from T1w MRI



Synthesizes fractional anisotropy maps from routine T1w MRI using tissue-aware modulation, prior-restoration decoding and adaptive fusion.

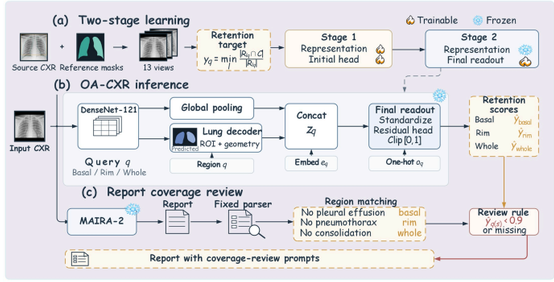
Image-level MAE 5.0% below ResViT; tract-level MAE and bias reduced by **20.2%** and **83.3%** across eight JHU tracts.

HCP Young Adult · 97 test subjects — **Paper**

03 REVIEW · First author · **Journal** Under review

OA-CXR

An Anatomical-Retention-Guided System for Selective Chest Radiograph Report Review



Estimates how much of each region a radiograph retains, then maps report statements to regional queries to flag those needing review.

Lowest error among nine retention regressors (macro MAE **0.0296**); reviewing 50% of reports captures **91.7%** of those with low-retention statements.

343,395 queries · 8,805 images · 4 datasets — **Paper** **Code**

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Status as of November 2026. Figures are method overviews from the respective papers. Personal research summary.